

Implementación de un SOC con ELK, Suricata y Beats

Monitorización de eventos de seguridad en endpoints Linux y Windows



Sumario

1. SIEM (Ubuntu 22.04 Desktop) — ELK + Logstash.....	3
1.1 Configuración ELK.....	3
1.2. Configuración de Logstash (SIEM).....	5
2. Suricata (IDS).....	9
3. Ingesta de datos (Beats).....	11
4. Linux endpoint: auth.log y Auditbeat.....	15
5. Windows endpoint: Sysmon + Winlogbeat.....	16

1. SIEM (Ubuntu 22.04 Desktop) — ELK + Logstash.

1.1 Configuración ELK.

Configuramos el repositorio de Docker y lo descargamos.

```
usuario@bash:~$ sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
[sudo] contraseña para usuario:
Des:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Des:2 http://es.archive.ubuntu.com/ubuntu jammy InRelease [270 kB]

usuario@bash:~$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF

usuario@bash:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
```

Creamos un single-node cluster por Docker y crearemos una nueva red Docker.

```
usuario@bash:~$ sudo docker network create elastic
74d296aa3062eaaa83738ff7e00cce4bb512b8c43555817f16d73191cb136bcb
```

Descargamos la imagen de Elasticsearch.

```
usuario@bash:~$ sudo docker pull docker.elastic.co/elasticsearch/elasticsearch:9.3.0
9.3.0: Pulling from elasticsearch/elasticsearch
d6e75bce595a: Download complete
4ca545ee6d5d: Download complete
1e745264966e: Download complete
884b2f340e03: Downloading [=====>] 122.7MB/695.7MB
317ce2aed5f4: Extracting 6 s
4b96446c3089: Download complete
4e734bfadf7d: Download complete
0e9b1e6d7391: Download complete
cd8469cd9638: Download complete
f59c6d9308d0: Download complete
```

Ajustamos el vm.max_map_count.

```
usuario@bash:~$ cat /proc/sys/vm/max_map_count
65530
usuario@bash:~$ sudo sysctl -w vm.max_map_count=262144
vm.max_map_count = 262144
usuario@bash:~$ echo "vm.max_map_count=262144" | sudo tee /etc/sysctl.d/99-elastic.conf
vm.max_map_count=262144
usuario@bash:~$ sudo sysctl --system
* Aplicando /etc/sysctl.d/10-console-messages.conf...
kernel.printk = 4 4 1 7
* Aplicando /etc/sysctl.d/10-ipv6-privacy.conf...
```

Creamos las carpetas donde tendremos el docker compose.

```
usuario@bash:~$ mkdir -p ~/soc-elk/logstash/pipeline
mkdir -p ~/soc-elk/logstash/config
cd ~/soc-elk
usuario@bash:~/soc-elk$ ls
logstash
```

Creamos el docker-compose.yml

services:

elasticsearch:

image: docker.elastic.co/elasticsearch/elasticsearch:8.15.0

container_name: soc-elasticsearch

restart: unless-stopped

environment:

- node.name=siem-es
- discovery.type=single-node
- xpack.security.enabled=false
- ES_JAVA_OPTS=-Xms512m -Xmx512m
- bootstrap.memory_lock=true

ulimits:

memlock:

soft: -1

hard: -1

nofile:

soft: 65535

hard: 65535

ports:

- "9200:9200"
- "9300:9300"

volumes:

- esdata:/usr/share/elasticsearch/data

kibana:

```
image: docker.elastic.co/kibana/kibana:8.15.0
container_name: soc-kibana
restart: unless-stopped
depends_on:
  - elasticsearch
environment:
  - SERVER_HOST=0.0.0.0
  - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
ports:
  - "5601:5601"
```

logstash:

```
image: docker.elastic.co/logstash/logstash:8.15.0
container_name: soc-logstash
restart: unless-stopped
depends_on:
  - elasticsearch
environment:
  - LS_JAVA_OPTS=-Xms256m -Xmx256m
ports:
  - "5044:5044"    # Beats
  - "9600:9600"    # API Logstash (opcional)
```

volumes:

```
- ./logstash/pipeline:/usr/share/logstash/pipeline:ro
-
```

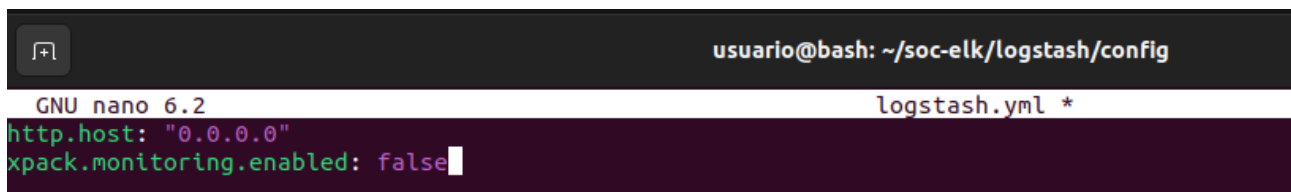
```
./logstash/config/logstash.yml:/usr/share/logstash/config/logstash
.yml:ro
```

volumes:

esdata:

1.2. Configuración de Logstash (SIEM).

Creamos el logstash.yml



```
usuario@bash: ~/soc-elk/logstash/config
GNU nano 6.2 logstash.yml *
http.host: "0.0.0.0"
xpack.monitoring.enabled: false
```

Y luego el 10-beats.conf.

```
input {
  beats {
    port => 5044
  }
}

filter {
  # GeoIP (para mapas en Kibana)
  if [source][ip] {
    geoip {
      source => "[source][ip]"
      target => "[source][geo]"
    }
  }
  if [destination][ip] {
    geoip {
      source => "[destination][ip]"
      target => "[destination][geo]"
    }
  }
}

output {
  # Si viene metadata de pipeline (módulos de Filebeat), úsala
```

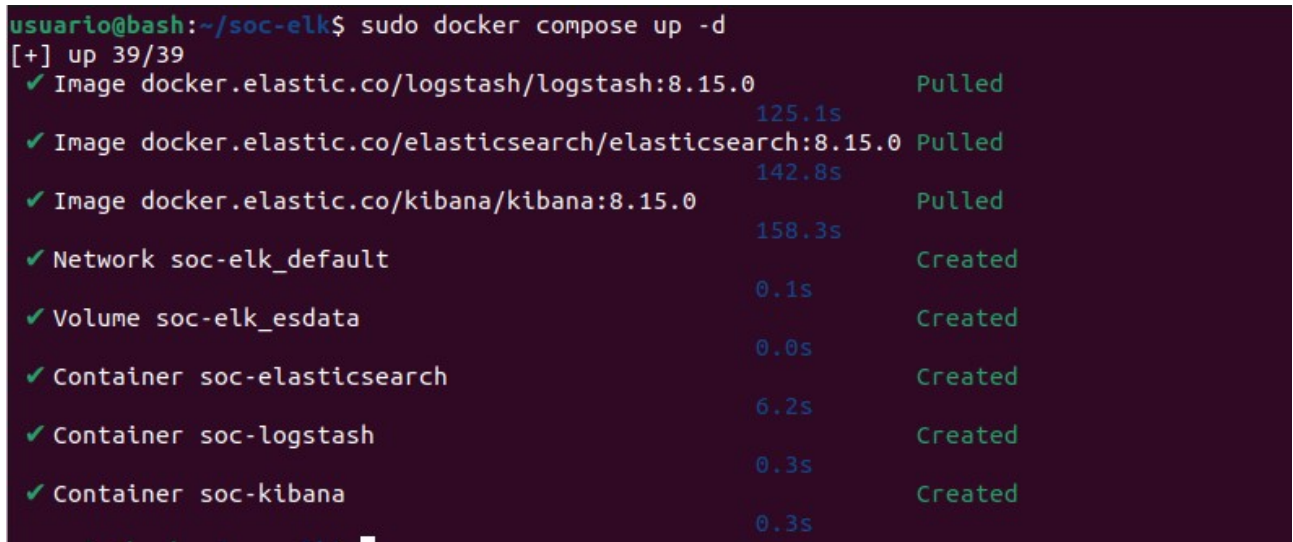
```

if [@metadata][pipeline] {
  elasticsearch {
    hosts => ["http://elasticsearch:9200"]
    index => "%{[@metadata][beat]}-%{+YYYY.MM.dd}"
    pipeline => "%{[@metadata][pipeline]}"
  }
} else {
  elasticsearch {
    hosts => ["http://elasticsearch:9200"]
    index => "%{[@metadata][beat]}-%{+YYYY.MM.dd}"
  }
}

# Debug (opcional; si molesta, comenta estas líneas)
# stdout { codec => rubydebug }
}

```

Levantamos el stack.



```

usuario@bash:~/soc-elk$ sudo docker compose up -d
[+] up 39/39
✓ Image docker.elastic.co/logstash/logstash:8.15.0 Pulled 125.1s
✓ Image docker.elastic.co/elasticsearch/elasticsearch:8.15.0 Pulled 142.8s
✓ Image docker.elastic.co/kibana/kibana:8.15.0 Pulled 158.3s
✓ Network soc-elk_default Created 0.1s
✓ Volume soc-elk_esdata Created 0.0s
✓ Container soc-elasticsearch Created 6.2s
✓ Container soc-logstash Created 0.3s
✓ Container soc-kibana Created 0.3s

```

Hacemos un curl para ver si responde elastic.

```

usuario@bash:~/soc-elk$ sudo curl http://localhost:9200
{
  "name" : "siem-es",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "s5vZQw37Sh-UHdYMRNuW5g",
  "version" : {
    "number" : "8.15.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "1a77947f34deddb41af25e6f0ddb8e830159c179",
    "build_date" : "2024-08-05T10:05:34.233336849Z",
    "build_snapshot" : false,
    "lucene_version" : "9.11.1",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}

```

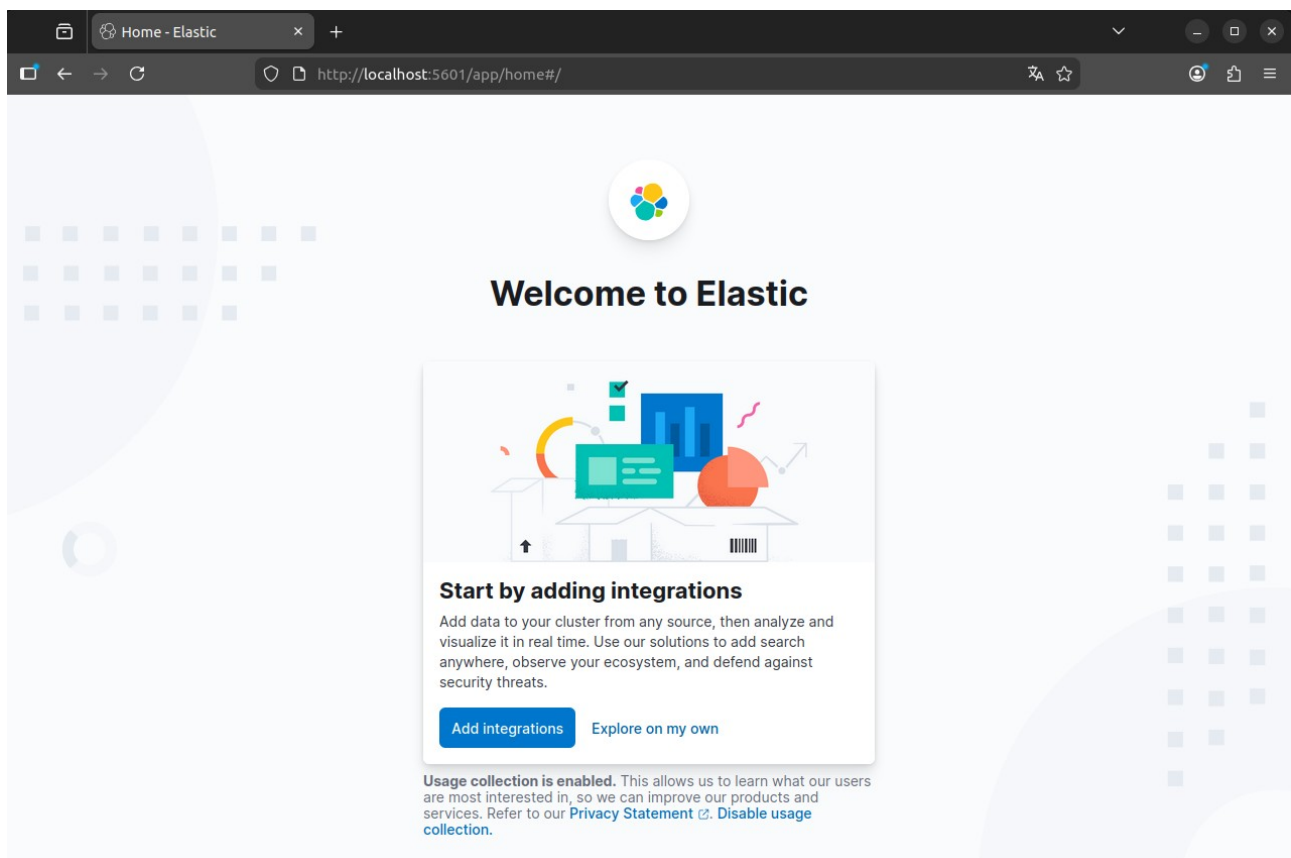
Y ver si logstash esta escuchando en su puerto por defecto que es el 5044.

```

usuario@bash:~/soc-elk$ ss -lntp | grep 5044
LISTEN 0      4096      0.0.0.0:5044  0.0.0.0:*
LISTEN 0      4096      [::]:5044  [::]:*

```

Y entramos en el navegador a ver si kibana funciona.



2. Suricata (IDS).

Actualizamos e instalamos suricata.

```
usuario@bash:~$ sudo apt-get install software-properties-common
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
software-properties-common ya está en su versión más reciente (0.99.22.9).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 136 no actualizados.
usuario@bash:~$ sudo add-apt-repository ppa:oisf/suricata-stable
Repositorio: «deb https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu/ jammy main»
Descripción:
Suricata IDS/IPS/NSM stable packages
https://suricata.io/
https://oisf.net/
```

```
usuario@bash:~$ sudo apt-get update
Obj:1 http://es.archive.ubuntu.com/ubuntu jammy InRelease
Obj:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Obj:3 http://es.archive.ubuntu.com/ubuntu jammy-updates InRelease
Obj:4 http://es.archive.ubuntu.com/ubuntu jammy-backports InRelease
Obj:5 https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu jammy InRelease
Leyendo lista de paquetes... Hecho
usuario@bash:~$ sudo apt-get install suricata
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
  libevent-core-2.1-7 libevent-pthreads-2.1-7 libhiredis0.14 libhyperscan5 liblua5.1-common libnet1 libnetfilter-queue1
Se instalarán los siguientes paquetes NUEVOS:
  libevent-core-2.1-7 libevent-pthreads-2.1-7 libhiredis0.14 libhyperscan5 liblua5.1-common libnet1 libnetfilter-queue1 suricata
0 actualizados, 8 nuevos se instalarán, 0 para eliminar y 348 no actualizados.
Se necesita descargar 7.206 kB de archivos.
Se utilizarán 33,7 MB de espacio de disco adicional después de esta operación.
¿Desea continuar? [S/n] s
```

Modificamos el suricata.yaml poniendo nuestra red e interfaz de red.

```
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"
```

```
# Linux high speed capture support
af-packet:
  - interface: enp0s3
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
```

Nos descargamos las reglas de ET Open:

```
usuario@bash:~$ sudo suricata-update
22/2/2026 -- 10:15:59 - <Info> -- Using data-directory /var/lib/suricata.
22/2/2026 -- 10:15:59 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
22/2/2026 -- 10:15:59 - <Info> -- Using /usr/share/suricata/rules for Suricata provided rules.
22/2/2026 -- 10:15:59 - <Info> -- Found Suricata version 8.0.3 at /usr/bin/suricata.
22/2/2026 -- 10:15:59 - <Info> -- Loading /etc/suricata/suricata.yaml
22/2/2026 -- 10:15:59 - <Info> -- Disabling rules for protocol pgsql
22/2/2026 -- 10:15:59 - <Info> -- Disabling rules for protocol modbus
22/2/2026 -- 10:15:59 - <Info> -- Disabling rules for protocol dnp3
22/2/2026 -- 10:15:59 - <Info> -- Disabling rules for protocol enip
22/2/2026 -- 10:15:59 - <Info> -- No sources configured, will use Emerging Threats Open
22/2/2026 -- 10:15:59 - <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-8.0.3/emerging.rules.tar.gz.
100% - 5360857/5360857
```

Reiniciamos Suricata.

```
usuario@bash:~$ sudo systemctl restart suricata
```

Para asegurarse de que Suricata se esté ejecutando, consultamos el registro de Suricata.

```
usuario@bash:~$ sudo tail -f /var/log/suricata/suricata.log
[5973 - Suricata-Main] 2026-02-22 10:17:23 Info: logopenfile: eve-log output device (regular) initialized: eve.json
[5973 - Suricata-Main] 2026-02-22 10:17:23 Info: logopenfile: stats output device (regular) initialized: stats.log
[5973 - Suricata-Main] 2026-02-22 10:17:48 Info: detect: 1 rule files processed. 48716 rules successfully loaded, 0 rules failed, 0 rules skipped
[5973 - Suricata-Main] 2026-02-22 10:17:48 Info: threshold-config: Threshold config parsed: 0 rule(s) found
[5973 - Suricata-Main] 2026-02-22 10:17:48 Info: detect: 48721 signatures processed. 1246 are IP-only rules, 4478 are inspecting packet payload, 42762 inspect application layer, 110 are decoder event only
[5973 - Suricata-Main] 2026-02-22 10:17:49 Notice: mpm-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
[5973 - Suricata-Main] 2026-02-22 10:17:49 Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
[5973 - Suricata-Main] 2026-02-22 10:17:49 Info: runmodes: enp0s3: creating 4 threads
[5975 - W#01-enp0s3] 2026-02-22 10:17:50 Info: ioctl: enp0s3: MTU 1500
[5973 - Suricata-Main] 2026-02-22 10:17:50 Notice: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.
```

Generamos una detección (ataque de prueba) y verificar alertas desde el Linux endpoint.

The screenshot shows a Linux desktop environment with three windows. The leftmost window is a LibreOffice Writer document titled 'Suricata'. The middle window is a terminal window titled 'usuario@bash:~\$' showing the output of the command 'sudo systemctl restart suricata' and the subsequent 'sudo tail -f /var/log/suricata/suricata.log' command. The terminal output shows the Suricata engine starting up, loading rules, and processing signatures. The rightmost window is another terminal window titled 'usuario@bash:~\$' showing the output of the command 'sudo tail -f /var/log/suricata/fast.log'. This window displays a series of ICMPv4 unknown code alerts from 192.168.0.20 to 192.168.0.30, indicating a network attack.

3. Ingesta de datos (Beats).

Intentamos instalar filebeat

```
usuario@bash:~$ sudo apt -y install filebeat
[sudo] contraseña para usuario:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
E: No se ha podido localizar el paquete filebeat
```

Pero como no lo encuentra, añadimos el repo de Elastic.

```
usuario@bash:~$ sudo apt -y install ca-certificates curl gnupg apt-transport-https
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
  dirmngr gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv libcurl4
```

```
usuario@bash:~$ curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elastic.gpg
```

```
usuario@bash:~$ echo "deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-8.x.list
deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main
```

Actualizamos e instalamos filebeat.

```
usuario@bash:~$ sudo apt update
Obj:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Obj:2 http://es.archive.ubuntu.com/ubuntu jammy InRelease
Des:3 https://artifacts.elastic.co/packages/8.x/apt stable InRelease [3.248 B]
Obj:4 http://es.archive.ubuntu.com/ubuntu jammy-updates InRelease
Obj:5 https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu jammy InRelease
Obj:6 http://es.archive.ubuntu.com/ubuntu jammy-backports InRelease
Des:7 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 Packages [100 kB]
Des:8 https://artifacts.elastic.co/packages/8.x/apt stable/main i386 Packages [3.396 B]
Descargados 107 kB en 1s (84,5 kB/s)
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se pueden actualizar 335 paquetes. Ejecute «apt list --upgradable» para verlos.
usuario@bash:~$ sudo apt -y install filebeat
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes NUEVOS:
  filebeat
0 actualizados, 1 nuevos se instalarán, 0 para eliminar y 335 no actualizados.
Se necesita descargar 68,4 MB de archivos.
Se utilizarán 263 MB de espacio de disco adicional después de esta operación.
Des:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 filebeat amd64 8.19.11 [68,4 MB]
Descargados 68,4 MB en 3s (24,4 MB/s)
Seleccionando el paquete filebeat previamente no seleccionado.
(Leyendo la base de datos ... 206565 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../filebeat_8.19.11_amd64.deb ...
Desempaquetando filebeat (8.19.11) ...
Configurando filebeat (8.19.11) ...
```

Activamos módulo de Suricata.

```
usuario@bash:~$ sudo filebeat modules enable suricata
Enabled suricata
```

Y comprobamos.

```
usuario@bash:~$ sudo filebeat modules list | grep suricata
suricata
```

Configurar el modulo suricata para activarlo.

```
usuario@bash: ~
GNU nano 6.2 /etc/filebeat/modules.d/suricata.yml *
# Module: suricata
# Docs: https://www.elastic.co/guide/en/beats/filebeat/8.19/filebeat-module-suricata.html

- module: suricata
  # All logs
  eve:
    enabled: true
    var.paths: ["/var/log/suricata/eve.json"]

  # Set custom paths for the log files. If left empty,
  # Filebeat will choose the paths depending on your OS.
  #var.paths:
```

Configurar salida a Logstash (SIEM) editando el filebeat.yml, en output.elasticsearch ponemos la ip de nuestra maquina con elastic.

```
usuario@bash: ~
GNU nano 6.2 /etc/filebeat/filebeat.yml *

# Kibana Space ID
# ID of the Kibana Space into which the dashboards should be loaded. By default,
# the Default Space will be used.
#space.id:

# ===== Elastic Cloud =====
# These settings simplify using Filebeat with the Elastic Cloud (https://cloud.elastic.co/).
# The cloud.id setting overwrites the `output.elasticsearch.hosts` and
# `setup.kibana.host` options.
# You can find the `cloud.id` in the Elastic Cloud web UI.
#cloud.id:

# The cloud.auth setting overwrites the `output.elasticsearch.username` and
# `output.elasticsearch.password` settings. The format is `<user>:<pass>`.
#cloud.auth:

# ===== Outputs =====
# Configure what output to use when sending the data collected by the beat.

# ----- Elasticsearch Output -----
output.elasticsearch:
  # Array of hosts to connect to.
  hosts: ["192.168.0.10:9200"]
```

Validar configuración y conexión.

```
usuario@bash:~$ sudo filebeat test config
Config OK
usuario@bash:~$ sudo filebeat test output
elasticsearch: http://192.168.0.10:9200...
  parse url... OK
  connection...
    parse host... OK
    dns lookup... OK
    addresses: 192.168.0.10
    dial up... OK
  TLS... WARN secure connection disabled
  talk to server... OK
  version: 8.15.0
```

Arrancar filebeat.


```

root@beatbox:~# sudo journalctl -u filebeat -n 50 --no-pager
Feb 22 11:03:37 bash systemd[1]: Started Filebeat sends log files to Logstash or directly to Elasticsearch..
Feb 22 11:03:37 bash filebeat[14860]: {"log.level": "info", "@timestamp": "2026-02-22T11:03:37.551+0100", "log.origin": {"function": "github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure", "file.name": "in
stance/beat.go", "file.line": 835}, "message": "Host path: [/usr/share/filebeat] Config path: [/etc/filebeat] Data path: [/var/lib/filebeat] Log path: [/var/log/filebeat], 'service.name': 'filebeat', 'ecs.version': '1
.6.0'"}
Feb 22 11:03:37 bash filebeat[14860]: {"log.level": "info", "@timestamp": "2026-02-22T11:03:37.551+0100", "log.origin": {"function": "github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure", "file.name": "in
stance/beat.go", "file.line": 843}, "message": "Beat ID: 88e43990-2457-4ea2-9173-93428df050f7, 'service.name': 'filebeat', 'ecs.version': '1.6.0'"}

```

Arranca Filebeat en la IDS y genera tráfico.

[illegible]

Desde la Debian endpoint (192.168.0.30) genera tráfico hacia la IDS.

```
usuario@bash:~$ ping -c 4 192.168.0.20
sudo nmap -sS -Pn -p 1-100 192.168.0.20
PING 192.168.0.20 (192.168.0.20) 56(84) bytes of data:
64 bytes from 192.168.0.20: icmp_seq=1 ttl=64 time=1.04 ms
64 bytes from 192.168.0.20: icmp_seq=2 ttl=64 time=0.649 ms
64 bytes from 192.168.0.20: icmp_seq=3 ttl=64 time=0.869 ms
64 bytes from 192.168.0.20: icmp_seq=4 ttl=64 time=2.33 ms

--- 192.168.0.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 0.649/1.222/2.328/0.653 ms
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-22 11:20 CET
Nmap scan report for 192.168.0.20
Host is up (0.0014s latency).
All 100 scanned ports on 192.168.0.20 are closed
MAC Address: 08:00:27:62:E3:D2 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

Comprueba en la SIEM que Logstash está recibiendo

```

root@warriorbash:~# ./src-elm5 ss -ntlp | grep 5044
LISTEN 0      4096        0.0.0.0:5044  0.0.0.0:*
LISTEN 0      4096        [::]:5044    [::]:*
root@warriorbash:~# ./src-elm5 docker logs soc-logstash -tail 100
permissions denied while trying to connect to the docker API at unix:///var/run/docker.sock
root@warriorbash:~# ./src-elm5 sudo docker logs soc-logstash -tail 100
[sudo] contraseña para usuario:
at io.netty.handler.codec.ByteToMessageDecoder.channelInactive(ByteToMessageDecoder.java:377) ~[netty-codec-4.1.109.Final.jar:4.1.109.Final]
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelInactive(AbstractChannelHandlerContext.java:383) ~[netty-transport-4.1.109.Final.jar:4.1.109.Final]
at io.netty.channel.AbstractChannelHandlerContext.access$353$0(AbstractChannelHandlerContext.java:414) ~[netty-transport-4.1.109.Final.jar:4.1.109.Final]
at io.netty.channel.AbstractChannelHandlerContext$54.run(AbstractChannelHandlerContext.java:286) ~[netty-transport-4.1.109.Final.jar:4.1.109.Final]
at io.netty.util.concurrent.AbstractEventExecutor.runTask(AbstractEventExecutor.java:173) ~[netty-common-4.1.109.Final.jar:4.1.109.Final]
at io.netty.util.concurrent.DefaultEventExecutor.run(DefaultEventExecutor.java:66) ~[netty-common-4.1.109.Final.jar:4.1.109.Final]
at io.netty.util.concurrent.DefaultEventExecutor$4.run(SingleThreadEventExecutor.java:997) ~[netty-common-4.1.109.Final.jar:4.1.109.Final]
at io.netty.util.internal.ThreadExecutorMap$2.run(ThreadExecutorMap.java:74) ~[netty-common-4.1.109.Final.jar:4.1.109.Final]
at io.netty.util.concurrent.FastThreadLocalRunnable.run(FastThreadLocalRunnable.java:30) ~[netty-common-4.1.109.Final.jar:4.1.109.Final]
at java.lang.Thread.run(Thread.java:583) ~[?:?]
Caused by: org.logstash.beats.InvaldFrameProtocolException: invalid version of beats protocol: 69
at org.logstash.beats.Protocol.version(Protocol.java:22) ~[logstash-input-beats-6.8.3.jar:~]
at org.logstash.beats.BeatsParser.decode(BeatsParser.java:62) ~[logstash-input-beats-6.8.3.jar:~]
at io.netty.handler.codec.ByteToMessageDecoder.decodeRemovalReentryProtection(ByteToMessageDecoder.java:530) ~[netty-codec-4.1.109.Final.jar:4.1.109.Final]
at io.netty.handler.codec.ByteToMessageDecoder.callDecode(ByteToMessageDecoder.java:469) ~[netty-codec-4.1.109.Final.jar:4.1.109.Final]
at io.netty.handler.codec.ByteToMessageDecoder.decode(ByteToMessageDecoder.java:469) ~[netty-codec-4.1.109.Final.jar:4.1.109.Final]

```

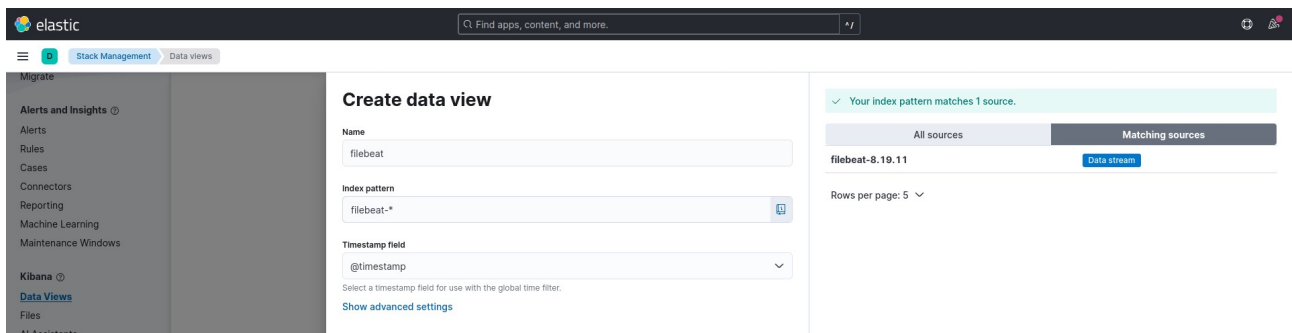
Verifica que Elasticsearch tiene índices filebeat-*

```

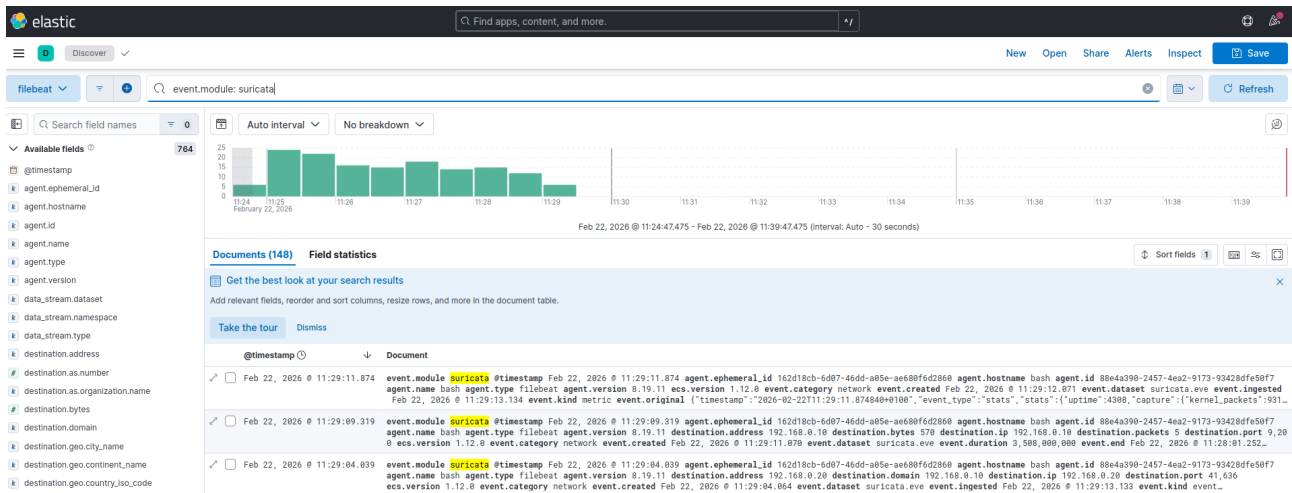
usuario@bash:~/soc-elk$ curl http://localhost:9200/_cat/indices?v | grep filebeat
 % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 2800    0 2800    0    0 10450    0 --:--:-- --:--:-- --:--:-- 10486
yellow open .ds-filebeat-8.19.11-2026.02.22-000001      ldjCXenrQaGHSIQ2F9yeag 1 1      6785      0      13mb

```

En Kibana: crea Data View para filebeat.



Ir a Discover y comprobar eventos de Suricata.



4. Linux endpoint: auth.log y Auditbeat.

Instalar Filebeat y Auditbeat en la linux endpoint, añadiendo la repo de Elastic.

```
usuario@bash:~$ sudo apt update
sudo apt -y install ca-certificates curl gnupg apt-transport-https
curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elastic.gpg
echo "deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-8.x.list
sudo apt update
[sudo] contraseña para usuario:
Obj:1 http://es.archive.ubuntu.com/ubuntu jammy InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Des:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
```

Instalar filebeat y auditbeat.

```
usuario@bash:~$ sudo apt -y install filebeat auditbeat
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes NUEVOS:
  auditbeat filebeat
0 actualizados, 2 nuevos se instalarán, 0 para eliminar y 95 no actualizados.
Se necesita descargar 115 MB de archivos.
Se utilizarán 443 MB de espacio de disco adicional después de esta operación.
Des:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 auditbeat amd64 8.19.11 [46,7 MB]
Des:2 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 filebeat amd64 8.19.11 [68,4 MB]
Descargados 115 MB en 12s (9.862 kB/s)
Seleccionando el paquete auditbeat previamente no seleccionado.
(Leyendo la base de datos ... 208014 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../auditbeat_8.19.11_amd64.deb ...
Desempaquetando auditbeat (8.19.11) ...
```

Filebeat para logs Linux (auth.log), activar módulo system.

```
usuario@bash:~$ sudo filebeat modules enable system
Enabled system
```

Configurar módulo system (auth), donde editamos system.yml

```
GNU nano 6.2 /etc/filebeat/modules.d/system.yml *
# Module: system
# Docs: https://www.elastic.co/guide/en/beats/filebeat/8.19/filebeat-module-system.html

module: system
# Syslog
syslog:
  enabled: true

# Set custom paths for the log files. If left empty,
# Filebeat will choose the paths depending on your OS.
#var.paths:

# Use journald to collect system logs
#var.use_journald: false

# Authorization logs
auth:
  enabled: true

# Set custom paths for the log files. If left empty,
# Filebeat will choose the paths depending on your OS.
#var.paths:

# Use journald to collect auth logs
#var.use_journald: false
```

Configurar salida a Logstash (SIEM) editando el filebeat.yml poniendo la IP de Elastic y comentando cualquier bloque output.elasticsearch.

```
GNU nano 6.2 /etc/filebeat/filebeat.yml
#cloud.id:

# The cloud.auth setting overwrites the `output.elasticsearch.username` and
# `output.elasticsearch.password` settings. The format is `<user>:<pass>`.
#cloud.auth:

# ===== Outputs =====

# Configure what output to use when sending the data collected by the beat.
# ----- Elasticsearch Output -----
#output.elasticsearch:
#  # Array of hosts to connect to.
#  # hosts: ["192.168.0.10:9200"]

#  # Performance preset - one of "balanced", "throughput", "scale",
#  # "latency", or "custom".
#  preset: balanced

#  # Protocol - either `http` (default) or `https`.
#  #protocol: "https"

#  # Authentication credentials - either API key or username/password.
#  #api_key: "id:api_key"
#  #username: "elastic"
#  #password: "changeme"

# ----- Logstash Output -----
output.logstash:
#  # The Logstash hosts
#  hosts: ["192.168.0.10:5044"]
```

Probar Filebeat.

```
usuario@bash:~$ sudo filebeat test config
Config OK
usuario@bash:~$ sudo filebeat test output
logstash: 192.168.0.10:5044...
connection...
  parse host... OK
  dns lookup... OK
  addresses: 192.168.0.10
  dial up... OK
TLS... WARN secure connection disabled
talk to server... OK
```

Arrancar Filebeat.


```

usuario@bash:~$ sudo systemctl enable --now filebeat
sudo systemctl status filebeat --no-pager
sudo journalctl -u filebeat -n 50 --no-pager
Created symlink /etc/systemd/system/multi-user.target.wants/filebeat.service → /lib/systemd/system/filebeat.service.
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
   Loaded: loaded (/lib/systemd/system/filebeat.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-02-22 18:34:00 CET; 79ms ago
     Docs: https://www.elastic.co/beats/filebeat
    Main PID: 8185 (filebeat)
      Tasks: 5 (limit: 2264)
     Memory: 23.2M
        CPU: 56ms
    CGroup: /system.slice/filebeat.service
            └─8185 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebeat.yml

Feb 22 18:34:00 bash systemd[1]: Started Filebeat sends log files to Logstash or directly to Elasticsearch..
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.564+0100","l
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.564+0100","l
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.609+0100","l
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.609+0100","l
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.609+0100","l
Feb 22 18:34:00 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.609+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.610+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.611+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.615+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.615+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"warn","@timestamp":"2026-02-22T18:34:00.615+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.616+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.665+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:00.935+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"warn","@timestamp":"2026-02-22T18:34:01.025+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:01.056+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:01.056+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:01.056+0100","l
Feb 22 18:34:01 bash filebeat[8185]: {"log.level":"info","@timestamp":"2026-02-22T18:34:01.270+0100","l
Hint: Some lines were ellipsized, use -l to show in full.
Feb 22 18:34:00 bash systemd[1]: Started Filebeat sends log files to Logstash or directly to Elasticsearch.

```

Generar eventos Linux (fallos de login) para el dashboard, para que aparezcan eventos en auth.log, genera algunos intentos fallidos.

```

usuario@bash:~$ sudo su probas
su: user probas does not exist or the user entry does not contain all the required fields
usuario@bash:~$ sudo su user
su: user user does not exist or the user entry does not contain all the required fields

```

Luego comprobar que se registró algo.

```

usuario@bash:~$ sudo tail -n 30 /var/log/auth.log
Feb 22 18:30:46 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:30:52 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/nano /etc/filebeat/filebeat.yml
Feb 22 18:30:52 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:31:08 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:31:11 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/nano /etc/filebeat/filebeat.yml
Feb 22 18:31:11 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:31:34 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:31:39 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/filebeat test config
Feb 22 18:31:39 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:31:39 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:31:43 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/filebeat test output
Feb 22 18:31:43 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:31:43 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:33:58 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/systemctl enable --now filebeat
Feb 22 18:33:58 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:34:00 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:34:00 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/systemctl status filebeat --no-pager
Feb 22 18:34:01 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:34:01 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:34:01 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/journalctl -u filebeat -n 50 --no-pager
Feb 22 18:34:01 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:34:01 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:42:22 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/su probas
Feb 22 18:42:22 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:42:22 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:42:37 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/su user
Feb 22 18:42:37 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Feb 22 18:42:37 bash sudo: pam_unix(sudo:session): session closed for user root
Feb 22 18:43:20 bash sudo: usuario : TTY=pts/0 ; PWD=/home/usuario ; USER=root ; COMMAND=/usr/bin/tail -n 30 /var/log/auth.log
Feb 22 18:43:20 bash sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)

```

Configurar auditbeat.yml.

```
usuario@bash: ~  
GNU nano 6.2 /etc/auditbeat/auditbeat.yml *  
## Unauthorized access attempts.  
#-a always,exit -F arch=b64 -S open,creat,truncate,ftruncate,openat,open_by_ho  
#-a always,exit -F arch=b64 -S open,creat,truncate,ftruncate,openat,open_by_ho  
  
- module: file_integrity  
  paths:  
    - /etc/passwd  
    - /etc/shadow  
  scan_at_start: true  
  
- module: system  
  datasets:  
    - login # User logins, logouts, and system boots.  
    - process # Started and stopped processes  
  
# ----- Logstash Output -----  
output.logstash:  
  # The Logstash hosts  
  hosts: ["192.168.0.10:5044"]
```

Probamos si la configuración está bien.

```
usuario@bash: ~  
usuario@bash:~$ sudo nano /etc/auditbeat/auditbeat.yml  
usuario@bash:~$ sudo auditbeat test config  
sudo auditbeat test output  
Config OK  
logstash: 192.168.0.10:5044...  
  connection...  
    parse host... OK  
    dns lookup... OK  
    addresses: 192.168.0.10  
    dial up... OK  
  TLS... WARN secure connection disabled  
  talk to server... OK
```

Si ambos salen bien, reiniciamos auditbeat y lo iniciamos.

```

usuario@bash:~$ sudo systemctl restart auditbeat
sudo systemctl enable auditbeat
sudo systemctl status auditbeat --no-pager
sudo journalctl -u auditbeat -n 50 --no-pager
Created symlink /etc/systemd/system/multi-user.target.wants/auditbeat.service → /lib/systemd/system/auditbeat.service.
● auditbeat.service - Audit the activities of users and processes on your system.
   Loaded: loaded (/lib/systemd/system/auditbeat.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-02-22 19:07:19 CET; 1s ago
     Docs: https://www.elastic.co/beats/auditbeat
   Main PID: 8511 (auditbeat)
      Tasks: 7 (limit: 2264)
     Memory: 35.0M
        CPU: 162ms
    CGroup: /system.slice/auditbeat.service
            └─8511 /usr/share/auditbeat/bin/auditbeat --environment systemd -c /etc/auditbeat/auditbea...

feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.957+0..._info"
feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.957+0...tem_in
feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.967+0100","...
feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.968+0100","...
feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.968+0... .6.0"}
feb 22 19:07:20 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:20.009+0100","...
feb 22 19:07:20 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:20.070+0100","...
feb 22 19:07:20 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:20.071+0100","...
feb 22 19:07:20 bash auditbeat[8511]: {"log.level":"warn","@timestamp":"2026-02-22T19:07:20.089+0100","...
feb 22 19:07:20 bash auditbeat[8511]: {"log.level":"warn","@timestamp":"2026-02-22T19:07:20.148+0100","...
Hint: Some lines were ellipsized, use -l to show in full.
feb 22 19:07:19 bash systemd[1]: Started Audit the activities of users and processes on your system..
feb 22 19:07:19 bash auditbeat[8511]: {"log.level":"info","@timestamp":"2026-02-22T19:07:19.839+0100","l
og.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":
"instance/beat.go","file.line":835},"message":"Home path: [/usr/share/auditbeat] Config path: [/etc/audi
tbeat] Data path: [/var/lib/auditbeat] Logs path: [/var/log/auditbeat]","service.name":"auditbeat","ecs.
version":"1.6.0"}

```

Creamos dataview de auditbeat-.*.

Create data view

Name:

Index pattern:

Timestamp field:

Select a timestamp field for use with the global time filter.

Show advanced settings

✓ Your index pattern matches 6 sources.

All sources	Matching sources
	auditbeat-2025.02.19 Index
	auditbeat-2025.03.15 Index
	auditbeat-2025.03.16 Index
	auditbeat-2025.03.17 Index
	auditbeat-2026.02.21 Index

Rows per page: 5 < 1 2 >

Generar evento de prueba para verlo en Kibana.

Discover - Elastic

http://localhost:5601/app/discover/#/?_g=()&refreshInterval=

Find apps, content, and more.

Discover

agent.type: "auditbeat"

Search field names

Available fields

@timestamp

@version

agent.ephemeral_id

agent.id

agent.name

agent.type

agent.version

auditd.data.acct

auditd.data.apparmor

auditd.data.class

auditd.data.cmd

auditd.data.denied_mask

auditd.data.grants

auditd.data.name

auditd.data.op

auditd.data.operation

Auto interval

No breakdown

Documents (376)

Field statistics

Get the best look at your search results

Add relevant fields, reorder and sort columns, resize rows, and more in the document table.

Take the tour

Dismiss

@timestamp

Document

Feb 22, 2026 @ 19:15:22.776

agent.type auditbeat @timestamp Feb 22, 2026 @ 19:15:22.776 @version 1 agent.ephemeral_id 8d5d8505-9760-4971-8279-5fbcd54aa6b agent.id b9e871b4-93b9-4f...

Feb 22, 2026 @ 19:15:17.683

agent.type auditbeat @timestamp Feb 22, 2026 @ 19:15:17.683 @version 1 agent.ephemeral_id 8d5d8505-9760-4971-8279-5fbcd54aa6b agent.id b9e871b4-93b9-4f...

usuario@bash: ~

usuario@bash:~\$ sudo touch /etc/passwd

usuario@bash:~\$

Discover

file.path: "/etc/passwd"

Search field names

Available fields

@timestamp

@version

agent.ephemeral_id

agent.id

agent.name

agent.type

agent.version

auditd.data.a0

auditd.data.a1

auditd.data.a2

auditd.data.a3

auditd.data.acct

auditd.data.apparmor

auditd.data.arch

auditd.data.class

auditd.data.cmd

Auto interval

No breakdown

Documents (2)

Field statistics

Get the best look at your search results

Add relevant fields, reorder and sort columns, resize rows, and more in the document table.

Take the tour

Dismiss

@timestamp

Document

Feb 22, 2026 @ 19:11:07.850

file.path /etc/passwd @timestamp Feb 22, 2026 @ 19:11:07.850 @version 1 agent.ephemeral_id 8d5d8505-9760-4971-8279-5fbcd54aa6b agent.id b9e871b4-93b9-4f...

Feb 22, 2026 @ 19:07:25.196

file.path /etc/passwd @timestamp Feb 22, 2026 @ 19:07:25.196 @version 1 agent.ephemeral_id 8d5d8505-9760-4971-8279-5fbcd54aa6b agent.id b9e871b4-93b9-4f...

Discover

event.module: "system"

Search field names

Available fields

@timestamp

@version

agent.ephemeral_id

agent.id

agent.name

agent.type

agent.version

auditd.data.a0

auditd.data.a1

auditd.data.a2

auditd.data.a3

auditd.data.acct

auditd.data.apparmor

auditd.data.arch

Auto interval

No breakdown

Documents (1)

Field statistics

Get the best look at your search results

Add relevant fields, reorder and sort columns, resize rows, and more in the document table.

Take the tour

Dismiss

@timestamp

Document

Feb 22, 2026 @ 19:23:22.664

event.module system @timestamp Feb 22, 2026 @ 19:23:22.664 @version 1 agent.ephemeral_id 8d5d8505-9760-4971-8279-5fbcd54aa6b agent.id b9e871b4-93b9-4f...

5. Windows endpoint: Sysmon + Winlogbeat.

Descargamos sysmon.

Learn / Sysinternals /

Sysmon v15.15

En este artículo

- Introducción
- Introducción a las funcionalidades de Sysmon
- Capturas de pantalla
- Uso
- Mostrar 5 más

Por Mark Russinovich y Thomas Garnier

Publicado: 23 de julio de 2024

 [Descargar Sysmon](#) (4,6 MB)

Ejecutamos en PowerShell.

```
PS C:\Users\paula\Downloads\Sysmon> .\Sysmon64.exe -accepteula -i

System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

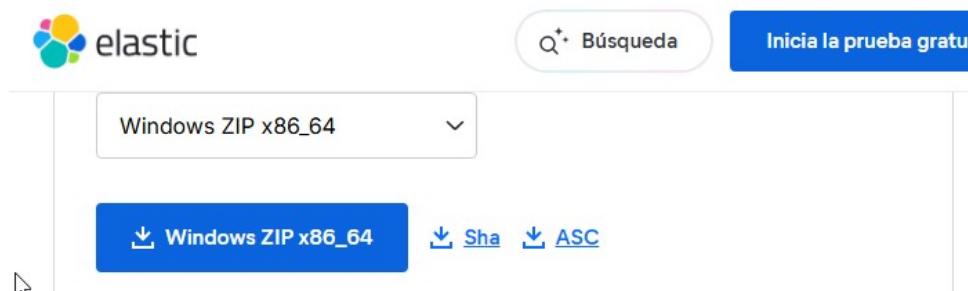
Comprobamos que existe el log.

```
PS C:\Users\paula\Downloads\Sysmon> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5

ProviderName: Microsoft-Windows-Sysmon

TimeCreated          Id LevelDisplayName Message
-----
22/02/2026 20:32:39   5 Información    Process terminated:...
22/02/2026 20:32:39   1 Información    Process Create:...
22/02/2026 20:32:29   5 Información    Process terminated:...
22/02/2026 20:32:22   5 Información    Process terminated:...
22/02/2026 20:32:22   1 Información    Process Create:...
```


Descargamos Winlogbeat.



Abre PowerShell como Administrador y edita el archivo winlogbeat.yml

```
winlogbeat.event_logs:
  - name: Application
    ignore_older: 72h

  - name: System

  - name: Security

  - name: Microsoft-Windows-Sysmon/Operational

# ----- Elasticsearch Output -----
#output.elasticsearch:
#  # Array of hosts to connect to.
#  #hosts: ["localhost:9200"]

#  # Protocol - either `http` (default) or `https`.
#  #protocol: "https"

#  # Authentication credentials - either API key or username/password.
#  #api_key: "id:api_key"
#  #username: "elastic"
#  #password: "changeme"

#  # Pipeline to route events to security, sysmon, or powershell pipelines.
#  pipeline: "winlogbeat-%[agent.version]}-routing"

# ----- Logstash Output -----
output.logstash:
  # The Logstash hosts
  hosts: ["192.168.0.10:5044"]
```



```
PS C:\Program Files\Winlogbeat> .\winlogbeat.exe test config -e
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.599+0100","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":836},"message":"Home path: [C:\\Program Files\\Winlogbeat] Config path: [C:\\Program Files\\Winlogbeat] Data path: [C:\\Program Files\\Winlogbeat\\data] Logs path: [C:\\Program Files\\Winlogbeat\\logs]","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.650+0100","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":844},"message":"Beat ID: 876e68f2-883a-4a06-b6b7-c89ade4d46f7","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.850+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).createBeater","file.name":"instance/beat.go","file.line":332},"message":"Setup Beat: winlogbeat; Version: 9.3.0 (FIPS-distribution: false)","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.901+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1393},"message":"Beat info","service.name":"winlogbeat","system_info":{"beat":{"path":{"config":"C:\\Program Files\\Winlogbeat","data":"C:\\Program Files\\Winlogbeat\\data","home":"C:\\Program Files\\Winlogbeat","logs":"C:\\Program Files\\Winlogbeat\\logs"},"type":"winlogbeat","uuid":"876e68f2-883a-4a06-b6b7-c89ade4d46f7"},"ecs.version":"1.6.0"}}}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.950+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1402},"message":"Build info","service.name":"winlogbeat","system_info":{"build":{"commit":"0f4Fc63162db855e0a1c5f0ec5894a8939e31d80"},"libbeat":"9.3.0","time":"2026-01-29T07:07:25.000Z","version":"9.3.0"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.956+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1405},"message":"Go runtime info","service.name":"winlogbeat","system_info":{"go":{"os":"windows","arch":"amd64","max_procs":2,"version":"go1.24.11"},"ecs.version":"1.6.0"}}}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.962+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1411},"message":"Host info","service.name":"winlogbeat","system_info":{"host":{"architecture":"x86_64","native_architecture":"x86_64","boot_time":"2026-02-22T19:13:57+01:00","name":"DESKTOP-CT3VFA4","ip":["192.168.0.40"],"mac":["08:00:27:6b:3f:e9"],"os":{"type":"windows","family":"windows","platform":"windows","name":"Windows 10 Pro"},"version":"10.0","major":10,"minor":0,"patch":0,"build":"19045.2965"},"timezone":"CET","timezone_offset_sec":3600,"id":"35ecab90-f21f-49f9-87ff-7d7fa617eac"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-02-22T20:00:42.972+0100","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1440},"message":"Process info","service.name":"winlogbeat","system_info":{"process":{"cwd":"C:\\Program Files\\Winlogbeat","exe":"C:\\Program Files\\Winlogbeat\\winlogbeat.exe","name":"winlogbeat.exe","pid":3860,"ppid":1864,"start_time":"2026-02-22T20:00:41.411+0100"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-02-22T20:00:43.174+0100","log.logger":"publisher","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/publisher/pipeline.LoadWithSettings","file.name":"pipeline/module.go","file.line":105},"message":"Beat name: DESKTOP-CT3VFA4","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-02-22T20:00:43.176+0100","log.logger":"winlogbeat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/publisher/pipeline.LoadWithSettings","file.name":"pipeline/module.go","file.line":105},"message":"State will be read from and persisted to C:\\Program Files\\Winlogbeat\\data\\winlogbeat.yml","service.name":"winlogbeat","ecs.version":"1.6.0"}
Config OK
```

```
PS C:\Program Files\Winlogbeat> .\winlogbeat.exe test output -e
{"log.level":"info", "@timestamp":"2026-02-22T20:12:56.490+0100", "log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Be
at).configure", "file.name":"instance/beat.go", "file.line":836}, "message":"Home path: [C:\\Program Files\\Winlogbeat] Config path: [C:\\Program
Files\\Winlogbeat] Data path: [C:\\Program Files\\Winlogbeat\\data] Logs path: [C:\\Program Files\\Winlogbeat\\logs]", "service.name":"winlogbea
t", "ecs.version":"1.6.0"}
{"log.level":"info", "@timestamp":"2026-02-22T20:12:57.094+0100", "log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Be
at).configure", "file.name":"instance/beat.go", "file.line":844}, "message":"Beat ID: 876e68f2-883a-4a06-b6b7-c89ade4d46f7", "service.name":"winlog
beat", "ecs.version":"1.6.0"}
logstash: 192.168.0.10:5044...
connection...
  parse host... OK
  dns lookup... OK
  addresses: 192.168.0.10
  dial up... OK
TLS... WARN secure connection disabled
talk to server... OK
```

```
PS C:\Program Files\Winlogbeat> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

Cambio de directiva de ejecución
La directiva de ejecución te ayuda a protegerte de scripts en los que no confías. Si cambias dicha directiva, podrías exponerte a los riesgos de seguridad descritos en el tema de la Ayuda about_Execution_Policies en https://go.microsoft.com/fwlink/?LinkID=135170. ¿Quieres cambiar la directiva de ejecución?
[S] Sí [O] Sí a todo [N] No [I] No a todo [U] Suspender [?] Ayuda (el valor predeterminado es "N"): s
PS C:\Program Files\Winlogbeat> .\install-service-winlogbeat.ps1

Status      Name            DisplayName
-----
Stopped     winlogbeat      winlogbeat
```

```
PS C:\Program Files\Winlogbeat> Start-Service winlogbeat
PS C:\Program Files\Winlogbeat> Get-Service winlogbeat
```

Status	Name	DisplayName
----	----	-----
Running	winlogbeat	winlogbeat

Creamos una data view en kibana para winlogbeat

Create data view

Name: winlogbeat

Index pattern: winlogbeat*

Timestamp field: @timestamp

Select a timestamp field for use with the global time filter.

Show advanced settings

✓ Your index pattern matches 2 sources.

All sources	Matching sources
winlogbeat-2026.02.21	Index
winlogbeat-2026.02.22	Index

Rows per page: 5

KQL para comprobar Winlogbeat.

Discover | agent.type: "winlogbeat"

Auto interval | No breakdown

Available fields: @timestamp, @version, agent.ephemeral_id, agent.id, agent.name, agent.type, agent.version, ecs.version, event.action, event.code, event.created, event.kind, event.original, event.outcome, event.provider, host.architecture, host.hostname, host.id, host.ip, host.mac

Documents (1,890) | Field statistics

Get the best look at your search results

Add relevant fields, reorder and sort columns, resize rows, and more in the document table.

Take the tour | Dismiss

@timestamp	Document
Feb 22, 2026 @ 20:53:48.682	agent.type winlogbeat @timestamp Feb 22, 2026 @ 20:53:48.682 @version 1 agent.ephemeral_id bca1d762-d880-429f-94fc-18fb828438c0 agent.id 182217e9-1869-4cd1-a55a-88b68ca5f33 agent.name DESKTOP-CT3VFA4 agent.version 9.3.0 ecs.version 8.0.0 event.action Process Create (rule: ProcessCreate) event.code 1 event.created Feb 22, 2026 @ 20:53:42.185 event.kind event event.original Process Create: RuleName: - UtcTime: 2026-02-22 19:53:48.680 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...
Feb 22, 2026 @ 20:53:48.682	agent.type winlogbeat @timestamp Feb 22, 2026 @ 20:53:48.682 @version 1 agent.ephemeral_id e6482328-2dff-4187-bafc-ebba8a4adc6 agent.id 876e68f2-883a-4a86-b6b7-c89ade4d46f7 agent.name DESKTOP-CT3VFA4 agent.version 9.3.0 ecs.version 8.0.0 event.action Process Create (rule: ProcessCreate) event.code 1 event.created Feb 22, 2026 @ 20:53:42.187 event.kind event event.original Process Create: RuleName: - UtcTime: 2026-02-22 19:53:48.680 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...
Feb 22, 2026 @ 20:53:48.598	agent.type winlogbeat @timestamp Feb 22, 2026 @ 20:53:48.598 @version 1 agent.ephemeral_id bca1d762-d880-429f-94fc-18fb828438c0 agent.id 182217e9-1869-4cd1-a55a-88b68ca5f33 agent.name DESKTOP-CT3VFA4 agent.version 9.3.0 ecs.version 8.0.0 event.action Logon event.code 4024 event.created Feb 22, 2026 @ 20:53:42.997 event.kind event event.original Se inició sesión correctamente en una cuenta. Firmante: Id. de seguridad: S-1-5-18 Nombre de cuenta: DESKTOP-CT3VFA4 Dominio de cuenta: WORKGROUP Id. de inicio de sesión: 6x3e7 Información de inicio de s...

Ver eventos de Sysmon.

Discover | winlog.channel: "Microsoft-Windows-Sysmon/Operational"

Auto interval | No breakdown

Available fields: @timestamp, @version, agent.ephemeral_id, agent.id, agent.name, agent.type, agent.version, ecs.version, event.action, event.code, event.created, event.kind, event.original, event.outcome, event.provider, host.architecture, host.hostname, host.id, host.ip, host.mac

Documents (650) | Field statistics

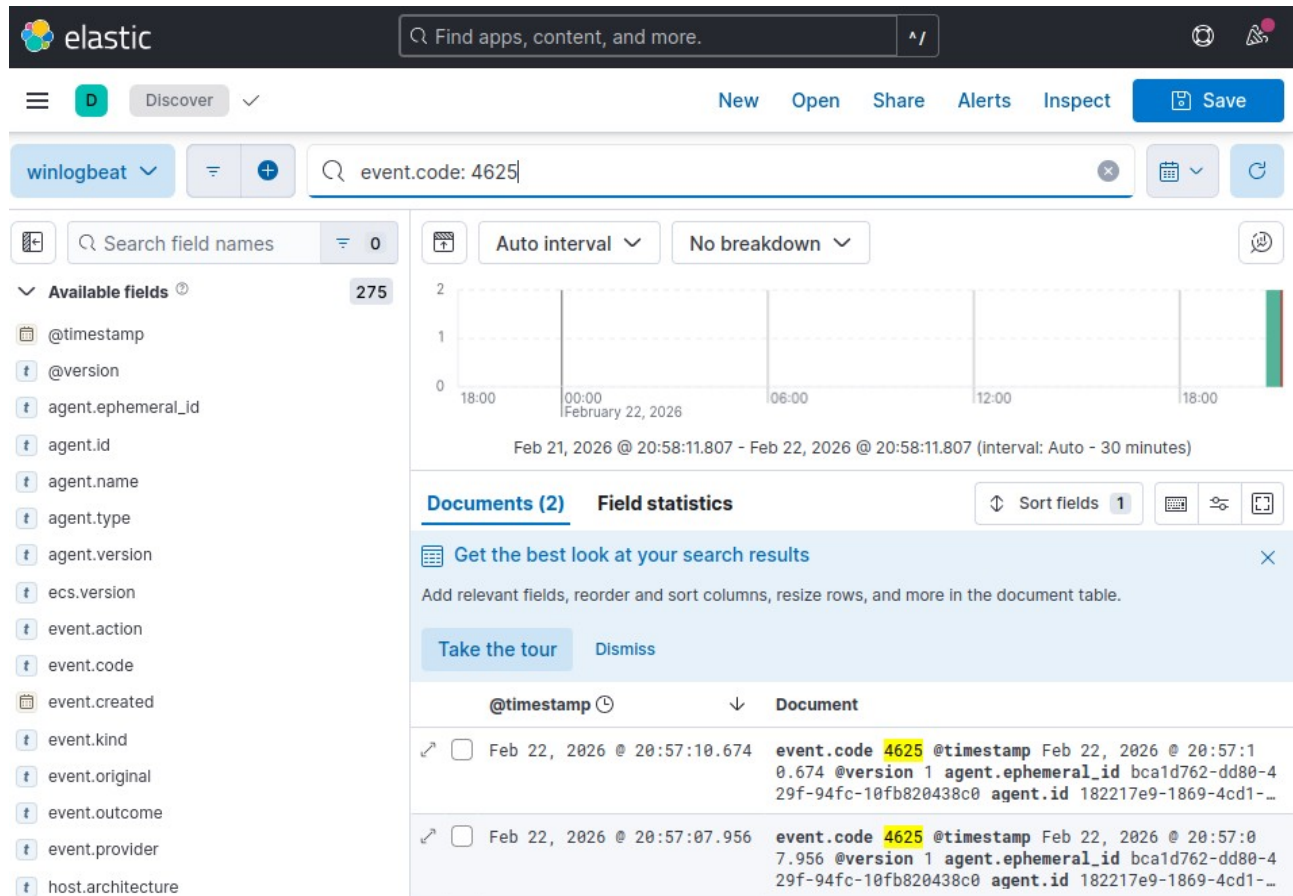
Get the best look at your search results

Add relevant fields, reorder and sort columns, resize rows, and more in the document table.

Take the tour | Dismiss

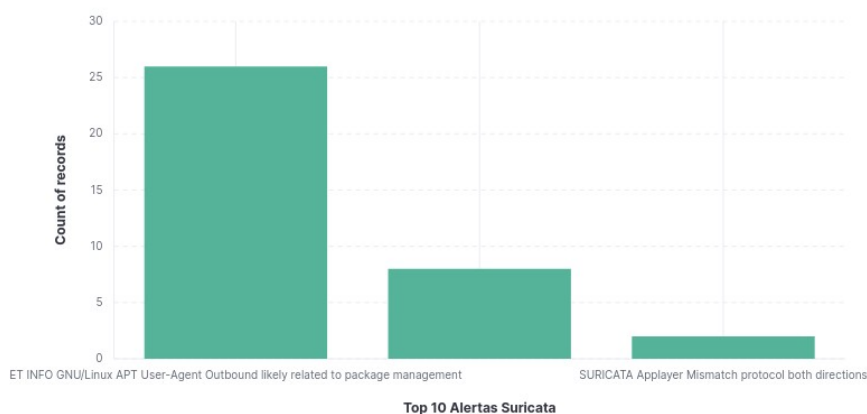
@timestamp	Document
Feb 22, 2026 @ 20:53:48.682	winlog.channel Microsoft-Windows-Sysmon/Operational @timestamp Feb 22, 2026 @ 20:53:48.682 @version 1 agent.ephemeral_id bca1d762-d880-429f-94fc-18fb828438c0 agent.id 182217e9-1869-4cd1-a55a-88b68ca5f33 agent.name DESKTOP-CT3VFA4 agent.type winlogbeat agent.version 9.3.0 ecs.version 8.0.0 event.action Process Create (rule: ProcessCreate) event.code 1 event.created Feb 22, 2026 @ 20:53:42.185 event.kind event event.original Process Create: RuleName: - UtcTime: 2026-02-22 19:53:48.680 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...
Feb 22, 2026 @ 20:53:48.682	winlog.channel Microsoft-Windows-Sysmon/Operational @timestamp Feb 22, 2026 @ 20:53:48.682 @version 1 agent.ephemeral_id e6482328-2dff-4187-bafc-ebba8a4adc6 agent.id 876e68f2-883a-4a86-b6b7-c89ade4d46f7 agent.name DESKTOP-CT3VFA4 agent.type winlogbeat agent.version 9.3.0 ecs.version 8.0.0 event.action Process Create (rule: ProcessCreate) event.code 1 event.created Feb 22, 2026 @ 20:53:42.187 event.kind event event.original Process Create: RuleName: - UtcTime: 2026-02-22 19:53:48.680 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...
Feb 22, 2026 @ 20:50:18.482	winlog.channel Microsoft-Windows-Sysmon/Operational @timestamp Feb 22, 2026 @ 20:50:18.482 @version 1 agent.ephemeral_id bca1d762-d880-429f-94fc-18fb828438c0 agent.id 182217e9-1869-4cd1-a55a-88b68ca5f33 agent.name DESKTOP-CT3VFA4 agent.type winlogbeat agent.version 9.3.0 ecs.version 8.0.0 event.action Process terminated (rule: ProcessTerminate) event.code 5 event.created Feb 22, 2026 @ 20:50:19.441 event.kind event event.original Process terminated: RuleName: - UtcTime: 2026-02-22 19:50:18.466 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...
Feb 22, 2026 @ 20:50:18.482	winlog.channel Microsoft-Windows-Sysmon/Operational @timestamp Feb 22, 2026 @ 20:50:18.482 @version 1 agent.ephemeral_id e6482328-2dff-4187-bafc-ebba8a4adc6 agent.id 876e68f2-883a-4a86-b6b7-c89ade4d46f7 agent.name DESKTOP-CT3VFA4 agent.type winlogbeat agent.version 9.3.0 ecs.version 8.0.0 event.action Process terminated (rule: ProcessTerminate) event.code 5 event.created Feb 22, 2026 @ 20:50:19.441 event.kind event event.original Process terminated: RuleName: - UtcTime: 2026-02-22 19:50:18.466 ProcessGuid: {35ECA890-5EC4-699B-7801-000000000000} ProcessId: 5892 Image: C:\Windows\System32\svchost.exe F...

Ver fallos de login Windows.

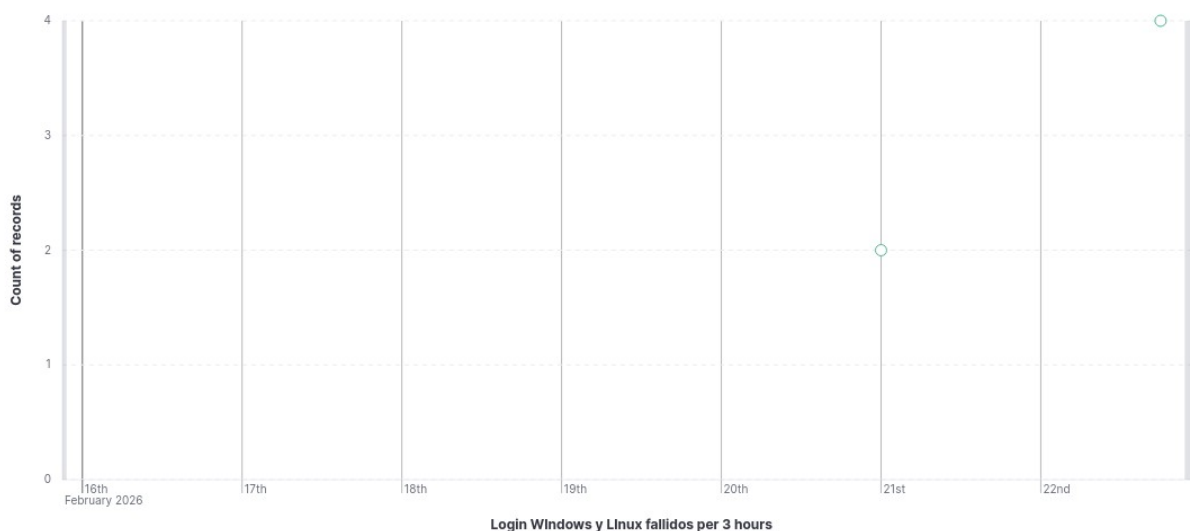


6. Crear un Dashboard personalizado que muestre.

- Mapa de geolocalización de IPs de origen (usando el filtro GeoIP).
GeoIP no muestra resultados en este laboratorio porque el tráfico capturado usa direcciones IP privadas (192.168.0.0/24), que no son geolocalizables. La capa GeoIP está preparada para funcionar con IPs públicas.
- Top 10 de alertas de Suricata más frecuentes.



- Gráfico de intentos de login fallidos en Windows y Linux.



- Configurar una regla de alerta simple (ej: detección de un escaneo de puertos Nmap).

Se intentó crear una regla de alerta simple en Kibana para detección de escaneos (por ejemplo, Nmap) sobre los eventos de Suricata (filebeat-*), con condición `count > 0` en una ventana de 5 minutos. La creación no pudo completarse en el laboratorio porque Kibana mostró el requisito de configuración adicional del sistema de alertas (“additional setup required”), normalmente relacionado con la configuración inicial de alerting/conectores/encryption key.